

VENKATESWARA REDDY SATTI

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Professional Summary

- Machine Learning Engineer with 4+ years of experience building production-grade ML systems, scalable AI infrastructure, distributed data pipelines, and low-latency backend services for high-volume real-time applications.
- Strong expertise in fraud detection, anomaly detection, behavioral modeling, recommendation systems, and conversational AI using Python, PyTorch, TensorFlow, Scikit-learn, and XGBoost.
- Hands-on experience designing and deploying end-to-end ML systems including feature engineering, distributed ETL pipelines, real-time inference APIs, model monitoring, experimentation, drift detection, and CI/CD workflows.
- Experienced in GenAI and LLM systems including LangChain, Retrieval-Augmented Generation (RAG), vector embeddings, vector databases, prompt engineering, and scalable AI inference services.
- Skilled in backend engineering and distributed systems using FastAPI, asynchronous processing, REST APIs, Docker, Kubernetes, MLflow, cloud-native services, and event-driven architectures.
- Proven ability to optimize ML pipelines and production inference systems for scalability, observability, reliability, throughput, and latency in cloud environments.

Education

Saint Louis University

St. Louis, MO

Master of Science in Artificial Intelligence — GPA: 3.8/4.0

2024 – 2025

- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Mining, Statistics, Distributed Systems

Professional Experience

Twilio

Atlanta, GA

Machine Learning Engineer – Real-Time ML Systems

June 2024 – Present

- Built production-grade real-time SMS spam and fraud detection systems using behavioral anomaly detection, ensemble learning, and deep learning models with PyTorch and Scikit-learn, improving messaging trustworthiness and reducing fraudulent traffic.
- Designed distributed event-driven ML pipelines for high-volume streaming data using Python, FastAPI, asynchronous processing, and cloud-native architectures to support low-latency real-time inference.
- Developed scalable conversational AI and GenAI systems using LangChain, Retrieval-Augmented Generation (RAG), embeddings, vector databases, and LLM orchestration pipelines for automated customer interactions.
- Engineered feature stores and feature engineering pipelines transforming large-scale user interaction data into production-ready ML features, improving model consistency, reliability, and inference quality.
- Built recommendation and behavioral modeling systems using reinforcement learning techniques to optimize message delivery timing, engagement scoring, and interaction personalization.
- Implemented model monitoring, observability, drift detection, and evaluation frameworks using MLflow and custom monitoring pipelines to ensure reliability and production model stability.
- Optimized distributed backend inference services using batching, caching, asynchronous APIs, and scalable microservices architectures, improving throughput by 30% and reducing end-to-end latency.
- Developed and deployed RESTful inference APIs and scalable backend microservices supporting real-time AI workloads and high-concurrency production environments.
- Collaborated with platform engineering and backend teams to integrate AI models into production systems using Docker, CI/CD pipelines, and cloud deployment workflows.
- Worked extensively with cloud infrastructure and distributed systems concepts including AWS services, containerized workloads, Kubernetes-based deployments, and scalable model-serving architectures.

- Built large-scale real-time transaction fraud detection systems using gradient boosting, anomaly detection, and ensemble learning models including XGBoost and Scikit-learn, improving fraud detection accuracy by 20%.
- Developed anti-money laundering (AML) detection systems using graph-based anomaly detection, behavioral analytics, and sequence modeling techniques to identify suspicious transaction behavior patterns.
- Designed credit risk and loan default prediction models using logistic regression, ensemble learning, and behavioral feature engineering pipelines for financial risk assessment.
- Built distributed ETL and feature engineering pipelines using Python, SQL, and asynchronous processing workflows for large-scale transactional datasets supporting production ML systems.
- Developed scalable real-time fraud scoring APIs and low-latency inference systems using FastAPI and distributed backend architectures, reducing fraud scoring latency by 25%.
- Implemented SHAP-based explainability, cross-validation workflows, and model evaluation pipelines to support compliance, transparency, and model governance requirements.
- Contributed to payment authorization optimization and intelligent transaction routing systems, improving approval rates and reducing revenue loss.
- Designed monitoring and retraining workflows for production fraud models including drift detection, performance monitoring, and automated experimentation pipelines.
- Collaborated with cross-functional engineering, risk, analytics, and platform teams to productionize scalable ML systems within high-throughput financial systems environments.
- Worked with cloud infrastructure, Docker-based deployments, CI/CD workflows, distributed databases, and production observability pipelines for scalable ML deployments.

Projects

- Built computer vision and instance segmentation systems using Mask R-CNN, DETR, CLIP, and PyTorch for large-scale object detection and segmentation workflows.
- Developed scalable end-to-end ML pipelines including distributed preprocessing, training orchestration, experimentation, evaluation, deployment, and monitoring workflows.
- Designed explainable AI and model optimization frameworks using LIME, SHAP, pruning strategies, reinforcement learning-based ranking systems, and domain-aware feature selection.
- Built AI-powered applications integrating FastAPI backend services, vector embeddings, Retrieval-Augmented Generation (RAG), and conversational AI workflows using LLM orchestration frameworks.
- Implemented Dockerized deployment pipelines, GitHub Actions CI/CD workflows, monitoring systems, and cloud-native inference services for scalable ML applications.

Technical Skills

- **Programming:** Python, SQL, Java, Bash/Shell Scripting
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Deep Learning, Reinforcement Learning, Fraud Detection, Anomaly Detection, Behavioral Modeling
- **GenAI / LLMOps:** LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, Prompt Engineering, Conversational AI, LLM Inference Pipelines
- **MLOps & ML Systems:** MLflow, CI/CD, Model Deployment, Drift Detection, Observability, Experiment Tracking, Monitoring, Model Serving
- **Data Engineering:** ETL Pipelines, Distributed Data Processing, Feature Engineering, Batch & Streaming Pipelines, Real-Time Processing
- **Backend & Distributed Systems:** FastAPI, REST APIs, AsyncIO, Microservices, Event-Driven Architectures, Distributed Systems
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda), Azure, GCP, Docker, Kubernetes, Cloud Deployment Workflows
- **Databases:** PostgreSQL, MySQL, MongoDB, Vector Databases
- **Tools & Platforms:** Git, GitHub Actions, Linux, Jupyter, Postman

Certifications

- Oracle Generative AI Professional
- DRDO Certified Artificial Intelligence Engineer